Q3LOCK Common-Alpha Topology and Critical Graph Route Split

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1 Purpose and scope

This note records EXP-000799 and strengthens reusable result R-167 to v1.3 without allocating a new result number. The exact result identifier remains PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT. The v1.0–v1.2 weighted-energy, cubic-multiplier, modular-topology and all-order-resummation results remain in force.

The new positive result is an exact all-bond unitary-kick theorem. The entire commuting cross-bond sector preserves a centered weighted energy graph up to a volume- and source-uniform factor and transfers canonical commutators by one lattice layer. At fixed finite volume this upgrades ordinary Lie–Trotter convergence to every subcritical graph power $s < 1/2$ once the standard strong product limit is supplied.

Three topology routes are also narrowed exactly. The all-bond kick is not point-norm continuous on a basic local resolvent. The quartic onsite flow destroys the unweighted and every subcritical $s < 1/2$ canonical Lipschitz class. A bounded coordinate cutoff does not yield a fixed-temperature half-modular-strip absolute expansion. A boundary-layer theorem further rejects every fixed Weyl-containing C-star-Leibniz critical seminorm that dominates a one-sided momentum commutator. The surviving primary obligation is therefore a non-Leibniz analytic/Fréchet or symmetric/state-weighted critical topology together with thermodynamic boundary Cauchy.

A separate finite-volume theorem shows that coordinate-tail truncations make the exact Gibbs state approximately invariant in trace norm at each fixed finite volume and fixed inverse temperature. A two-level fixture proves that small direct $D, \delta D$ differences do not imply uniform evolved half-strip multipliers. A faithful-representation fixture proves that representationwise strong-star convergence is not an abstract C-star conclusion.

No common thermodynamic automorphism, common-alpha KMS identification, algebraic ground states, broken-sector GNS or physical mass gap, regulator or continuum removal, physical empty-space comparison, C6, CP1, Sector A or Pre-A closure is claimed.

2 Provenance identifiers

- Result: PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT, v1.3, R-167.
- Exploration: EXP-000799.
- Active gate: PA-CP1-ST8-Q3LOCK-ALL-BOND-UNITARY-TROTTER-GRAPH-LIPSCHITZ-AND-COMMON-ALPHA-CLOSURE.
- Active gate: PA-CP1-ST8-Q3LOCK-PROJECTED-DUHAMEL-MODULAR-C1-MULTIPLIER-LOCALITY.
- Negative: NG-2026-08-10-PRE-A-ST8-Q3LOCK-RAW-LOCAL-RESOLVENT-POINT-NORM-BOND-KICK-CONTINUITY.

- Negative: NG-2026-08-10-PRE-A-ST8-Q3LOCK-UNWEIGHTED-ONSITE-QP-LIPSCHITZ-STABILITY.
- Negative: NG-2026-08-10-PRE-A-ST8-Q3LOCK-SUBCRITICAL-ENERGY-DAMPED-ONSITE-LIPSCHITZ-STABILITY.
- Negative: NG-2026-08-10-PRE-A-ST8-Q3LOCK-COORDINATE-CUTOFF-HALF-MODULAR-STRIP-ABSOLUTE-CLOSURE.
- Negative: NG-2026-08-10-PRE-A-ST8-Q3LOCK-SMALL-D-DELTA-D-UNIFORM-HALF-STRIP-MULTIPLIER-INFERENCE.
- Negative: NG-2026-08-10-PRE-A-ST8-Q3LOCK-FAITHFUL-REPRESENTATION-STRONGSTAR-ABSTRACT-CSTAR-INFERENCE.
- Negative: NG-2026-08-10-PRE-A-ST8-Q3LOCK-CRITICAL-ONE-SIDED-ENERGY-DAMPED-LEIBNIZ-ONSITE-STABILITY.

3 Exact all-bond kick

At finite volume write

$$K_X = \sum_x f_x k_x, \qquad k_x \geq 1 + \frac{p_x^2}{2\chi} + \gamma |q_x|^4, \tag{3.1}$$

where $e^{-\mu} \leq f_x/f_y \leq e^{\mu}$ for neighbours. Split the Hamiltonian as an onsite tensor sum plus

$$V_{\text{cross}} = -c \sum_{\langle xy \rangle} q_x \cdot q_y, \qquad B_\delta = e^{-i\delta V_{\text{cross}}/\hbar}. \tag{3.2}$$

Every cross-bond factor commutes with every other, even when two bonds share a vertex. The exact canonical action is

$$B_\delta^* q_x B_\delta = q_x, \qquad B_\delta^* p_x B_\delta = p_x + \delta c S_x, \qquad S_x = \sum_{y \sim x} q_y. \tag{3.3}$$

For lattice degree z,

$$\sum_x f_x |S_x|^2 \leq z^2 e^{\mu} \sum_y f_y |q_y|^2 \leq \frac{z^2 e^{\mu}}{2\sqrt{\gamma}} K_X. \tag{3.4}$$

Expanding the shifted kinetic square and using Young's inequality in both time orientations gives, for $|\delta| \leq 1$,

$$B_{\pm\delta}^* K_X B_{\pm\delta} \leq (1 + C_b|\delta|) K_X, \qquad C_b = 1 + \frac{c^2 z^2 e^{\mu}}{2\chi\sqrt{\gamma}}. \tag{3.5}$$

Heinz interpolation therefore yields

$$\|K_X^s B_{\pm\delta} K_X^{-s}\| \leq (1 + C_b|\delta|)^s, \qquad 0 \leq s \leq \frac{1}{2}. \tag{3.6}$$

For $c = 3/5$, $z = 6$, $\chi = 7/4$, $\sqrt{\gamma} = 2/5$ and $e^{\mu} = 3/2$, the exact constant is $C_b = 521/35$. Each orientation in $N_s(C) = \|CK_X^{-s}\| + \|K_X^{-s}C\|$ costs the exact power s; a fully conjugated $K_X^s A K_X^{-s}$ estimate may safely cost $2s$ because it uses both kick factors.

4 One-layer recurrence and subcritical Trotter theorem

With $\beta_\delta(A) = B_\delta^* A B_\delta$, the CCR give

$$[q_x, \beta_\delta(A)] = \beta_\delta([q_x, A]), \tag{4.1}$$

$$[p_x, \beta_\delta(A)] = \beta_\delta\left([p_x, A] - \delta c \sum_{y\sim x} [q_y, A]\right). \tag{4.2}$$

Thus the bond step transfers influence by one lattice layer and one explicit power of δ. The onsite tensor flow is a K_X graph isometry. Suppose the finite-volume Lie–Trotter products $T_N(t)$ converge strongly to $T(t)$ and, on $|t| \le T_0$,

$$\sup_N \|K_X^{1/2} T_N(\pm t) K_X^{-1/2}\| \le C_T. \tag{4.3}$$

Fix $0 \le s < 1/2$. For $\xi \in D(K_X^{1/2-s})$, set $\eta_N = (T_N - T)K_X^{-s}\xi$. Then $\eta_N \to 0$ in norm, $\|K_X^{1/2}\eta_N\| \le 2C_T \|K_X^{1/2-s}\xi\|$, and

$$\|K_X^s \eta_N\| \le \|\eta_N\|^{1-2s} \|K_X^{1/2}\eta_N\|^{2s} \longrightarrow 0. \tag{4.4}$$

The uniform conjugated bounds extend this convergence by density from $D(K_X^{1/2-s})$ to all of $D(K_X^s)$. The argument does not close the endpoint $s = 1/2$ and gives no thermodynamic boundary Cauchy theorem.

5 Raw resolvent topology fails

On one bond and one component let $R_x = (i + p_x)^{-1}$. The convention in (3.3) gives

$$\beta_\delta(R_x) = (i + p_x + \delta c q_y)^{-1}. \tag{5.1}$$

The strongly commuting pair (p_x, q_y) has joint spectrum $\mathbb{R}^2$. Hence, for every nonzero δ,

$$\|\beta_\delta(R_x) - R_x\| = \sup_{u,v\in\mathbb{R}} \frac{|u - v|}{\sqrt{1+u^2}\sqrt{1+v^2}} = 1, \tag{5.2}$$

because

$$(1+u^2)(1+v^2) - (u-v)^2 = (1+uv)^2 \tag{5.3}$$

and $uv = -1$ saturates the bound. This rejects point-norm continuity of the kick on the raw basic-resolvent topology. It does not reject the fixed kick as an automorphism, a weaker graph or strict topology, normal W-star dynamics, or the full Hamiltonian flow.

6 Quartic onsite obstruction and the critical boundary

The one-component submodel already suffices:

$$h = \frac{p^2}{2\chi} + \frac{gq^4}{4}, \qquad W_a = e^{-iap/\hbar}. \tag{6.1}$$

Although $[p, W_a] = 0$, differentiation on the Schwartz core gives

$$\left.\frac{d}{dt}\right|_{t=0}[p, \alpha_t(W_a)] = g(3aq^2 - 3a^2q + a^3)W_a. \tag{6.2}$$

The strong core derivative is an unbounded quadratic multiplier, so an ordinary bounded canonical Lipschitz estimate cannot hold. On translated compact bumps, $K \sim \gamma R^4$ and $q^2K^{-s} \sim R^{2-4s}$. Both one-sided repairs therefore fail for every $s < 1/2$. The scalar power count is neutral at $s = 1/2$, but the fixed Leibniz endpoint also fails.

6.1 Critical one-sided Leibniz no-go

For the full one-site Q3LOCK onsite Hamiltonian set $K = h - \inf \sigma(h) + 1$, $W_a = e^{-iap_0/\hbar}$ and $t_a = \tau/a^2$. The exact Q3 force is

$$\partial_0 V_4 = (g + 3\lambda)q_0^3 - \frac{3\lambda}{2}q_0^2 \sum_{j\sim 0} q_j + \lambda q_0 \sum_{j\sim 0} q_j^2 - \frac{\lambda}{2}\sum_{j\sim 0} q_j^3. \tag{6.3}$$

Quartic coercivity bounds the translated energy by $E_* a^4$. During $0 \le s \le \tau/a^2$, evolved momenta are therefore $O(a^2)$ and every translated coordinate displacement from ae_0 remains $O_\tau(1)$ in L^2. Substitution in (6.3) yields

$$\langle \partial_0 V\rangle_s = (g + 3\lambda)a^3 + O_\tau(a^2). \tag{6.4}$$

Integration of the exact Heisenberg equation, followed by a ground-vector matrix element, gives the operator lower bound

$$\|[p_0, \alpha_{t_a}(W_a)]K^{-1/2}\| \ge (g + 3\lambda)\tau a - B_\tau, \tag{6.5}$$

with B_τ independent of a. If a fixed star-symmetric C-star-Leibniz seminorm L is finite on one nonzero W_b, dominates either one-sided critical p_0 commutator and obeys $L(\alpha_t(A)) \le (1 + C|t|)L(A)$, then $L(W_b^n) \le nL(W_b)$. Taking $a = nb$ in (6.5) and a fixed $\tau > L(W_b)/(c_p(g + 3\lambda)b)$ gives contradictory linear slopes while $t_a \to 0$.

For $g = 3/5$, $\lambda = 2/7$, $\chi = 7/4$, the independently reconstructed Q3 jets are

$$G = \frac{51}{35}, \qquad Dp_0 = \frac{51}{35}a^3, qquad D^2p_0 = 0, \qquad D^3p_0 = -\frac{32112}{8575}a^5. \tag{6.6}$$

The result rejects only the fixed one-sided-dominating C-star-Leibniz route. Non-Leibniz analytic/Fréchet, symmetric or state-weighted, and direct projected topologies remain open.

7 Coordinate-cutoff half-strip boundary

For $Q_L(q) = \eta(|q|/L)q$, the cutoff bond is bounded, but

$$[p^2, Q_L(q)] = -i\hbar\{p \cdot DQ_L + DQ_L \cdot p\} \tag{7.1}$$

is unbounded. Thus boundedness alone does not make the cutoff norm-C^1 for the quartic onsite derivation. Even granting an auxiliary analytic repair, a degree-z connected absolute expansion with bond norm J_L has parameter

$$r = \frac{2zJ_L|\operatorname{Im} t|}{\hbar}. \tag{7.2}$$

At the half modular strip, $|\operatorname{Im} t| = \beta\hbar/2$, this requires $z\beta J_L < 1$. Since $J_L = \Theta(cL^2)$, no fixed positive β survives $L \to \infty$ through this absolute method. This is a route boundary, not a no-dynamics theorem and not a rejection of direct projected $D, \delta D$.

8 Fixed-volume relative-unitary theorem

At fixed finite Λ and fixed β, let

$$\rho = Z^{-1}e^{-\beta H}, \qquad K = H - W_L, \qquad \rho_t = U_K \rho U_K^*. \tag{8.1}$$

For bounded W_L, Duhamel in its two orientations gives

$$\|(U_K - U_H)\rho^{1/2}\|_2, \ \|\rho^{1/2}(U_K - U_H)\|_2 \le \frac{|t|}{\hbar}\phi(W_L^2)^{1/2}. \tag{8.2}$$

Consequently

$$\|U_K \rho U_K^* - \rho\|_1 \le \frac{2|t|}{\hbar}\phi(W_L^2)^{1/2}. \tag{8.3}$$

The unbounded coordinate tail follows by form approximation. On the common quartic form domain,

$$|W_L| \le cz \sum_x |q_x|^2, \qquad |W_L - W_{L,M}| \le cz|\Lambda|M^{-2}\sum_x |q_x|^4. \tag{8.4}$$

Thus bounded outer cutoffs converge in form norm, the corresponding Hamiltonians converge in strong resolvent sense, and their unitaries converge strongly on compact real-time intervals. Gaussian domination gives both Hilbert–Schmidt tail orientations. If $E_L = \phi(W_L^2)^{1/2} \to 0$, then

$$\sup_{|t|\le T} \|U_K \rho U_K^* - \rho\|_1 \to 0 \quad (L \to \infty) \tag{8.5}$$

for each fixed finite Λ and fixed β. The constants may depend on both.

Finite Gibbs energy and K-energy conservation also give

$$S(\rho_t\|\rho) = \beta[\rho_t(W_L) - \rho(W_L)]. \tag{8.6}$$

If $\phi(e^{\theta[W_L - \phi(W_L)]}) < \infty$ for some $\theta > \beta$, the Gibbs variational inequality and Golden–Thompson yield

$$S(\rho_t\|\rho) \le \frac{\beta}{\theta - \beta}\log\phi(e^{\theta[W_L - \phi(W_L)]}). \tag{8.7}$$

This is not uniform in volume or inverse temperature and does not construct a separating common local algebra.

9 Direct topology is weaker than half-strip multiplier control

Let

$$H_n = \mathrm{diag}(0, n), \quad \epsilon_n = e^{-\beta n/4}, \quad K_n = H_n + \epsilon_n \sigma_x, \quad A = P_0. \tag{9.1}$$

At $t_n = \pi\hbar/\sqrt{n^2 + 4\epsilon_n^2}$, the perturbation, its first modular derivative, the direct difference $D_n = \tau_{t_n}^{K_n}(P_0) - P_0$, and $[\log \rho_n, D_n]$ all tend to zero in Duhamel norm. However, the off-diagonal entry implies

$$M_0(\tau_{t_n}^{K_n}(P_0)) \gtrsim \frac{2e^{\beta n/4}}{n} \longrightarrow \infty, \tag{9.2}$$

and M_1 diverges faster. Therefore direct projected $D, \delta D$ locality is strictly weaker than uniform evolved half-strip multiplier control; the two must not be called equivalent.

10 Representation topology boundary

Let $\mathcal{A} = \ell^\infty$ and $f_n = 1_{\{k \geq n\}}$. In the faithful multiplication representation on ℓ^2, $f_n \to 0$ strong-star. Adjoining a nonprincipal-ultrafilter character leaves the direct-sum representation faithful but changes the limit to $0 \oplus 1$. Thus a strong-star limit in one faithful representation does not determine an abstract C-star limit. The projected route must preregister one faithful/separating state class and may at most construct a fixed-temperature common normal W-star system until an independent phase- and beta-independent algebra theorem is supplied.

11 Route decision

The fixed critical C-star-Leibniz route is rejected. The all-bond route can continue only through a non-Leibniz analytic/Fréchet or symmetric/ state-weighted critical topology, followed by a volume-independent boundary Cauchy estimate and an invariant Hamiltonian algebra. The projected route remains secondary: prove direct $D, \delta D$ Cauchy on a preregistered separating local test class, product closure, exhaustion independence and the group law.

12 Devil's-advocate review

The package survived the following hostile checks.

1. The resolvent sign was recomputed from the declared convention $\beta_\delta = B_\delta^*(\cdot)B_\delta$; the plus sign in (5.1) is required.
2. The one-sided seminorm cost is M_δ^s, while M_δ^{2s} is only a safe fully conjugated bound. They are not conflated.
3. The Trotter graph conclusion is extended from $D(K_X^{1/2-s})$ by a uniform conjugated bound and density; the endpoint is not inferred.
4. Neutral scalar power counting at $s = 1/2$ does not rescue the fixed Leibniz route: the exact boundary layer (6.5) defeats its linear Weyl-power growth. Non-Leibniz and state-weighted alternatives are not rejected.
5. The unbounded coordinate tail uses a common closed form domain, form-norm cutoff convergence, finite Gibbs energy and an exponential moment; no thermodynamic uniformity is inferred.
6. The raw-resolvent, onsite and cutoff negatives reject named proof topologies or methods only. None is promoted to nonexistence of exact dynamics.

Therefore v1.3 is a verified topology and route split. It is not a proof of common alpha, KMS phase identity, ground-state selection, a GNS gap, continuum physics, physical empty space, C6, Sector A or Pre-A.

13 Reproduction

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_common_alpha_topology_critical_
graph_route_split.py

E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_common_alpha_topology_critical_
graph_route_split_independent.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_common_alpha_topology_critical_
graph_route_split_verify.py
```

14 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167 v1.3; no new result number.
Exploration:	EXP-000799.
Precise statement:	All-bond kick graph form and one-layer recurrence proved; subcritical finite-volume graph Trotter convergence proved conditionally on strong product convergence. Raw resolvent, unweighted/subcritical onsite Lipschitz, fixed critical Leibniz, cutoff half-strip, multiplier inference and abstract-C-star shortcuts rejected.
Scope:	T0 exact finite-volume topology theorems and scoped proof-route counterexamples.
Dependencies:	EXP-000781, EXP-000790, EXP-000794--EXP-000799; R-167 v1.0--v1.2.
Evidence grade:	ANALYTIC + EXACT EXECUTED + INDEPENDENT + PDF QA.
Reproduction command:	Run the three commands under Reproduction in the listed order, with the integrated verifier last.
Expected output:	Primary PASS 77/77; independent PASS 111/111; integrated PASS 271/271.
Falsification gate:	Any CCR sign, weighted graph factor, interpolation domain, resolvent norm, onsite exponent, half-strip radius, form cutoff, topology, hash or scope mismatch.
Tier before / after:	C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:	Non-Leibniz/state-weighted critical onsite closure, thermodynamic Cauchy, common alpha/KMS, algebraic ground states, GNS gap, regulator removal, continuum, physical empty and the below-empty sign, C6, CP1, Sector A and Pre-A remain open.
Next required action:	Prove a non-Leibniz or state-weighted critical onsite topology and thermodynamic boundary Cauchy, or close direct projected D,delta-D locality on a separating preregistered class.